Discrete event simulation

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Basic concepts

System A collection of entities that act and interact in order to achieve a certain logical goal.
The state of a system is the collection of variables necessary to describe the system at a particular moment.

Model A simplified description of a system, aimed at evaluating its performance or the effect of certain decisions.

Simulation Evolving the model of a system by providing appropriate inputs, then observing and analyzing the results.

Two main system types: discrete and continuous.

Models

Physical vs mathematical model.

Analytical model, numerical model, model with simulation.

Reference textbook: A. M. Law, Simulation Modeling and Analysis, fifth edition, McGraw-Hill, USA, 2015.

Event approach

Assumption: system in which state variables can only change at a countable number of points in time.

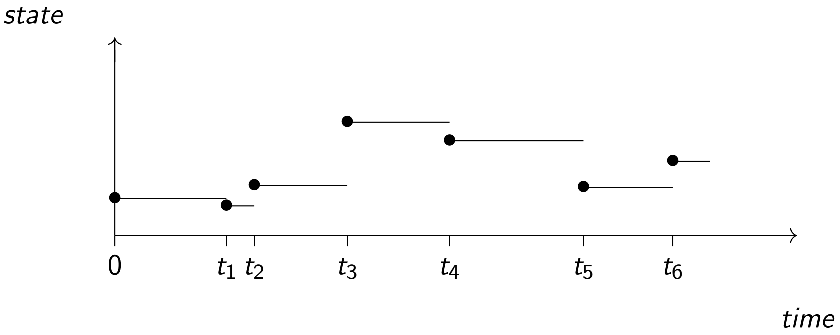
Sequence of events $e_0, e_1, e_2 \ldots$, occurring at times $0 \leq t_0 \leq t_1 \leq t_2 \leq \ldots$.

$\mathcal{S}_i$: system state immediately after e_i.

Simulation time: current value of t_i.

$(t_i, \mathcal{S}_i)$ must contain enough information to continue the simulation (except the realizations of the random variables for events e_j, $j > i$).

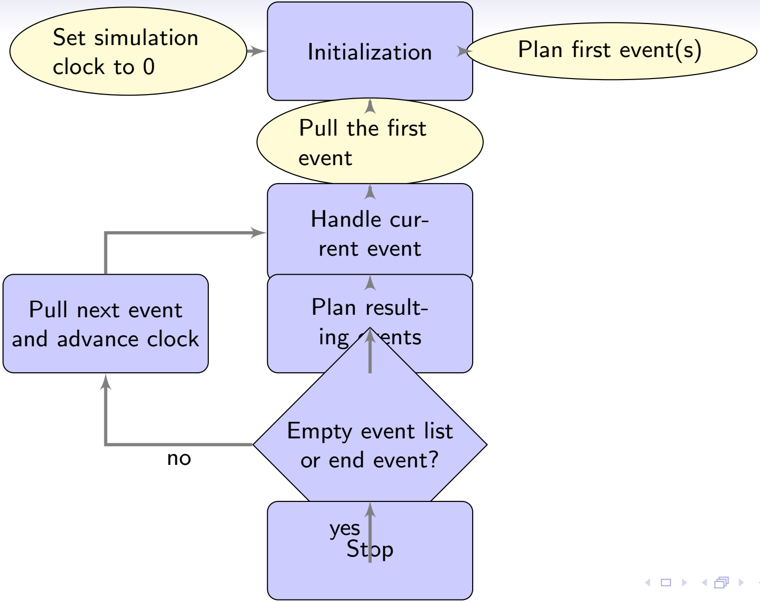
Discrete events

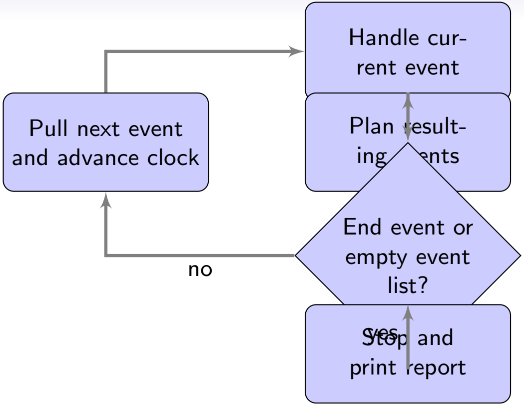


General framework

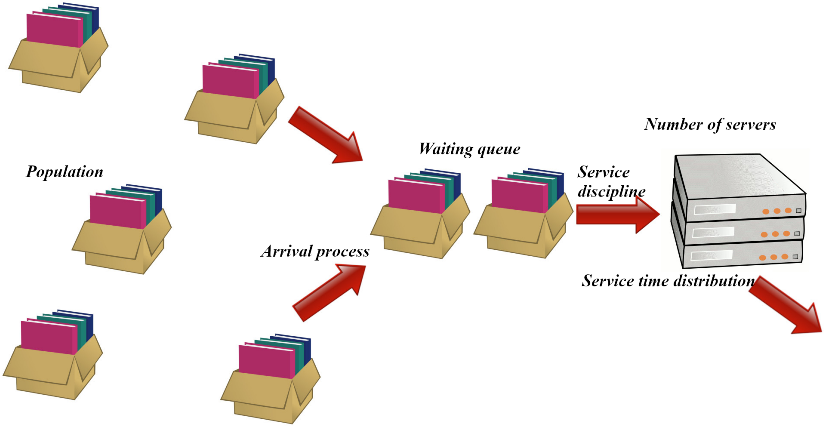
- Main program.
- Simulation clock.
- Event list.
- Random number generation functions.
- Statistical collectors.
- Report generation.

Event: action, schedule, cancel.


Set simulation clock to 0
Initialization
Plan first event(s)
Pull the first event
Handle current event
Pull next event and advance clock
Plan resulting events
Empty event list or end event?
no
yes
Stop


Handle cur-
rent event
Pull next event
and advance clock
Plan result-
End event or
empty event
list?
no
yes
Stop and
print report

Queueing systems



Kendall notation

$$A/S/c/K/N/SD$$

- A Arrival process
- S Service time distribution
- c Number of servers
- K Waiting list maximum length
- N Population size
- SD Service discipline

Notation

G General

I Independent

M Markovian or memoryless

Defaults:

- Infinite buffer capacity
- Infinite population size
- FCFS (First-Come-First-Served) service discipline

Memoryless property:

$$P[X > x + a \,|\, X > a] = P[X > x]$$

Continuous: exponential distribution
Discrete: geometric distribution

Example: GI/G/1 queue

$$G/G/1 = G/G/1/\infty/\infty/FCFS$$

We also write $GI/G/1$ to stress that the service times are independent.

The clients arrive one by one, one being served at a time, under a FCFS priority, with

- S_i: service time of client i, with cdf G;
- A_i: inter-arrival time between clients i and $i+1$, with cdf F.

We assume S_i and A_i mutually independent.

Example: GI/G/1 queue

The first client arrives at time A_0 in an empty system.

We wish to simulate this system for a duration T and calculate the average waiting time per client and the average queue length. The types of events are

1. Arrival;
2. Departure;
3. End of simulation.

The random variables (independent) to generate are $A_1, A_2, A_3, \ldots$ and $S_1, S_2, S_3, \ldots$.

Example: GI/G/1 queue

- W_i, the waiting time of client i;
- $Q(t)$, the length of the queue at time t;
- $N_c(t)$, the number of clients who started their service during the interval $[0, t]$.

Suppose we want to calculate, for the interval $[0, T]$, the *average waiting time* per client,

$$\overline{W}_{N_c(T)} = \frac{1}{N_c(T)} \sum_{i=1}^{N_c(T)} W_i;$$

and the *average queue length*:

$$\overline{Q}_T = \frac{1}{T} \int_0^T Q(t) dt.$$

Example: GI/G/1 queue

This last integral is easy to calculate through simulation.

For each *client*, we create an *object* containing the arrival time and service duration.

State variables: list of clients waiting and the list of clients in service.

$+$ simulation clock, statistical counters (if needed), *list* of future scheduled events, in chronological order, and a procedure for each type of event.

Queueing System Simulation

Arrival Generate A according to F and schedule another arrival in A time units;
Create the new client and note their arrival time;
Generate their service duration S according to G;
If (server is busy) **then**
    Insert this client into the waiting list;
else
    Insert the client into the service list;
    Schedule their departure in S time units;
Update the desired statistics.

Departure Remove the client from the service list;
If the queue is not empty **then**
    Remove the first client from the waiting list;
    Insert them into the service list;
    Schedule their departure in S time units;
Update the desired statistics.

End of Simulation Print a report and terminate the program.

Simulation

Start the simulation:

- Initialize variables and counters,
- Schedule the "End of Simulation" at time T,
- Generate A according to F and schedule the first "Arrival" in A time units,
- Launch the simulation.

The execution of the simulation simply consists of repeating the following loop:
Repeat: execute the next event in the event list
until: the list of scheduled events is empty,
or an event stops the simulation.

Simulation without Event List

It is not always necessary to resort to the event-driven approach.

Example: Lindley's recurrence:

$$W_1 = 0, \quad W_{i+1} = \max(0,\ W_i + S_i - A_{i+1}).$$

We can easily simulate for a fixed number of clients (instead of a fixed time horizon T).

The program formulation based on Lindley's recurrence reduces to solving an integration problem over $[0,1)^t$.

Lindley's Recurrence

Suppose we want to estimate $E[\overline{W}_{100}]$ by $\overline{W}_{100}$:

$$\overline{W}_{100} = \frac{1}{100}\sum_{i=1}^{100} W_i.$$

We need $A_1, S_1, A_2, S_2, \ldots, A_{99}, S_{99}$. If we set $A_i = F^{-1}(U_{2i-1})$ and $S_i = G^{-1}(U_{2i})$, where U_j, $j = 1, \ldots, 99$, are uniform random variables over $(0,1)$, then $\overline{W}_{100} = f(U_1, \ldots, U_{198})$ (with f being an unknown function in this case).

We then have $t = 198$, i.e.,

$$E[\overline{W}_{100}] = \int_{[0,1)^{198}} f(\boldsymbol{u})d\boldsymbol{u}.$$

Random Number of Clients

On the other hand, if we want to simulate $\overline{W}_{N_c(T)}$, the number of clients $N_c(T)$ is random and unbounded. We should therefore choose $t = \infty$ as the dimension.

Process Approach

A process is a temporally ordered sequence of interrelated events, separated by certain time intervals, that describes the entire experience of an “entity” as it evolves through a “system”. A simulation system or model may involve different types of processes.

A routine will be associated with each process in the model, describing its entire history through the system.

In contrast to the event-driven approach, a process routine explicitly contains the passage of simulated time and generally has multiple entry points.

Processes vs Events

A simulation using the process approach also evolves over time by executing events in the order of their occurrence.

Internally, the event-driven and process approaches are very similar.

For the user, reasoning may differ, and it can sometimes (but not always) feel more natural to use the process approach.

Improving Efficiency

The following material is adapted from Pierre L'Ecuyer's class "Simulation: stochastic aspects".

The efficiency of an estimator X is defined as follows:

$$eff[X] = \frac{1}{MSE[X]C(X)},$$

where MSE stands for mean square error and $C(X)$ represents the cost to compute X.

Is it possible, given X, to construct a new estimator Y that is more efficient than X?

We will introduce the main concepts of improving efficiency using an introductory example about call centers.

Introductory Example

Let B be the *load factor* for the day, and suppose that $P[B = b_t] = q_t$, where

t	1	2	3	4
b_t	0.8	1.0	1.2	1.4
q_t	0.25	0.55	0.15	0.05

It is easy to verify that $E[B] = 1$.

The *arrivals* of calls follow a Poisson process with rate $B\lambda_j$ during hour j.

Let $G_i(s)$ be the number of calls that started service after less than s seconds of waiting on day i.

We want to estimate $\mu = E[G_i(s)]$, say for $s = 20$.

Example

We also assume that the service durations of the calls follow the distribution $\Gamma(\alpha, \gamma)$, with a mean of $\alpha\gamma$.

In our example, we have $\alpha = 1$ and $\gamma = \gamma_1 = 100$.

Number of agents n_j and arrival rate λ_j (per hour) for 13 hourly periods in the call center:

j	0	1	2	3	4	5	6	7	8	9	10	11	12
n_j	4	6	8	8	8	7	8	8	6	6	4	4	4
λ_j	100	150	150	180	200	150	150	150	120	100	80	70	60

We simulate n independent days.

Example

Let $X_i = G_i(s)$ for day i, and

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

We have $E[\bar{X}_n] = \mu$ and $Var[\bar{X}_n] = Var[X_i]/n$.

An experiment with $n = 1000$ yields $\bar{X}_n = 1518.3$ and $S_n^2 = 21615$. The estimated variance of $\bar{X}_n$ is then $\widehat{Var}[\bar{X}_n] = 21.6$.

Introductory Example

We want to construct a confidence interval for $\sigma^2 = nVar[\bar{X}_n]$, under the assumption that $(n-1)S_n^2/\sigma^2$ approximately follows a χ^2 distribution with $n-1$ degrees of freedom. This allows us to construct the 90% interval: $[0.930S_n^2, 1.077S_n^2]$.

In other words, the relative error of this estimator is less than 8% with a "confidence" of approximately 90%.

Let's see how to *improve* this estimator $\bar{X}_n$ by reducing its variance. For each proposed method, we will provide numerical results for $n = 1000$.

Indirect Estimation

Let A_i be the total number of arrivals on day i; let $D_i = A_i - X_i$.

We know that $a = E[A_i] = \sum_{j=1}^{m} \lambda_j = 1660$.

Thus, we can write $\mu = E[X_i] = E[A_i - D_i] = a - E[D_i]$, which can be estimated by

$$\bar{X}_{\mathrm{i,n}} = E[A_i] - \bar{D}_n = a - \frac{1}{n}\sum_{i=1}^{n} D_i.$$

This estimator has less variance than $\bar{X}_n$ if and only if $Var[D_i] < Var[X_i]$. Estimated variance: $\widehat{Var}[X_{\mathrm{i,i}}] = 18389$.

Control Variate (CV)

Idea: leverage auxiliary information.

For example, if A_i is higher than usual ($A_i > E[A_i] = 1660$), we expect that on that day, X_i and D_i *overestimate* $E[X_i]$ and $E[D_i]$.

We could make a "*correction*" to these estimators: replace X_i with

$$X_{\mathrm{c,i}} = X_i - \beta(A_i - 1660)$$

where β is an appropriate constant. Then

$$\bar{X}_{\mathrm{c,n}} = \bar{X}_n - \beta(\bar{A}_n - 1660).$$

Control Variate (CV)

We have $E[\bar{X}_{\mathrm{c,n}}] = E[X_i]$ and

$$Var[\bar{X}_{\mathrm{c,n}}] = \frac{Var[X_i] + \beta^2 Var[A_i] - 2\beta Cov[A_i, X_i]}{n}.$$

This variance is a quadratic function of β, which we minimize by taking

$$\beta = \beta^* = \frac{Cov[A_i, X_i]}{Var[A_i]}.$$

The *minimum variance* is

$$Var[\bar{X}_{\mathrm{c,n}}] = \frac{Var[X_i] - (\beta^*)^2 Var[A_i]}{n} = Var[\bar{X}_n](1 - \rho^2[A_i, X_i])$$

where $\rho[A_i, X_i]$ is the correlation coefficient between A_i and X_i.

Control Variate (CV)

We do not know $Cov[A_i, X_i]$, but:

(a) We can estimate it through *pilot experiments*.

(b) We can estimate it using the *same n simulations* as $\bar{X}_n$.

With (b), we obtain the estimator (slightly biased):

$$\bar{X}_{\mathrm{c,e,n}} = \bar{X}_n - \frac{1}{n-1}\left[\sum_{i=1}^{n}(A_i - \bar{A}_n)(X_i - \bar{X}_n)\right]\frac{\bar{A}_n - a}{Var[A_i]}.$$

Conditional on B_i, $A_i \sim \text{Poisson}(1660 B_i)$. Thus, we have

$$\begin{aligned} Var[A_i] &= Var[E[A_i|B_i]] + E[Var[A_i|B_i]] \\ &= Var[1660 B_i] + E[1660 B_i] \\ &= 1660^2 Var[B_i] + 1660 E[B_i] \\ &= 67794.4. \end{aligned}$$

The empirical variance obtained here is: 3310.

Control Variate (CV)

By taking $\beta = 1$, we recover the indirect estimator:

$$\bar{X}_{\mathrm{i,n}} = \bar{X}_n - (\bar{A}_n - 1660) = 1660 - \bar{D}_n.$$

If we combine *CV* + *indirect*, we obtain:

$$\begin{aligned} \bar{X}_{\mathrm{i,c,n}} &= a - \bar{D}_n - \beta_2(\bar{A}_n - a) \\ &= \bar{A}_n - \bar{D}_n - (1+\beta_2)(\bar{A}_n - a) \\ &= \bar{X}_n - (1+\beta_2)(\bar{A}_n - a), \end{aligned}$$

i.e., $\bar{X}_{\mathrm{i,c,n}}$ is equivalent to $\bar{X}_{\mathrm{c,n}}$ with $\beta = 1 + \beta_2$. As a result, in the presence of the control variable, the indirect estimation provides no additional benefit.

We could also consider other control variables, such as B_i, the average service durations, etc.

Stratification

The load factor B_i is a significant source of variability in this case. Let's try to control it.

By setting $\mu_t = E[X_i \mid B_i = b_t]$, we have

$$\begin{aligned}
\mu &= E[X_i] = \sum_{t=1}^{4} P[B_i = b_t] \cdot E[X_i \mid B_i = b_t] \\
&= 0.25\, E[X_i \mid B_i = 0.8] + 0.55\, E[X_i \mid B_i = 1.0] \\
&\quad +0.15\, E[X_i \mid B_i = 1.2] + 0.05\, E[X_i \mid B_i = 1.4] \\
&= 0.25\, \mu_1 + 0.55\, \mu_2 + 0.15\, \mu_3 + 0.05\, \mu_4.
\end{aligned}$$

Idea: estimate μ_t *separately* for each t.

Stratification

Suppose there are N_t days where $B_i = b_t$ and let $X_{t,1}, \ldots, X_{t,N_t}$ be the values of X_i for those days.

We can estimate $\mu_t = E[X_i \mid B_i = b_t]$ by

$$\hat{\mu}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} X_{t,i}$$

and μ by

$$\bar{X}_{\mathrm{s,n}} = \sum_{t=1}^{4} q_t \hat{\mu}_t = 0.25\hat{\mu}_1 + 0.55\hat{\mu}_2 + 0.15\hat{\mu}_3 + 0.05\hat{\mu}_4.$$

Stratification

We have

$$
\begin{aligned}
&Var[\bar{X}_{\mathrm{s,n}} \mid N_1, N_2, N_3, N_4] \\
&= \sum_{t=1}^{4} q_t^2 \, Var[\hat{\mu}_t | N_t] = \sum_{t=1}^{4} q_t^2 \frac{\sigma_t^2}{N_t} \\
&= 0.25^2 \frac{\sigma_1^2}{N_1} + 0.55^2 \frac{\sigma_2^2}{N_2} + 0.15^2 \frac{\sigma_3^2}{N_3} + 0.05^2 \frac{\sigma_4^2}{N_4}.
\end{aligned}
$$

where $\sigma_t^2 = Var[X_i \mid B_i = b_t]$.
The variance is reduced if the σ_t^2 are less than $Var[X_i]$.
If the B_i are normally generated: *post-stratification*.
To estimate μ through stratification, we can also *fix the $N_t = n_t$ in advance*, meaning we choose in advance how many days will have $B_i = b_t$ for each value of t.

Stratification

- *Proportional allocation*: set $n_t = nq_t$.
  With $n = 1000$, this gives $n_1 = 250$, $n_2 = 550$, $n_3 = 150$, $n_4 = 50$.
- *Optimal allocation*: choose the n_t to minimize $Var[\bar{X}_{\mathrm{s,n}}]$ under the constraint $n_1 + n_2 + n_3 + n_4 = n$. We obtain:

$$\frac{n_t}{n} = \frac{\sigma_t P[B_i = t]}{\sum_{k=1}^{4} \sigma_k P[B_i = k]} = \frac{\sigma_k q_k}{\sum_{k=1}^{4} \sigma_k q_k}.$$

  We do not know these σ_k, but we can estimate them through pilot trials. With $n_0 = 800$ pilot trials, 200 for each value of t, we obtain, for example, $(n_1, n_2, n_3, n_4) = (219, 512, 182, 87)$ (after rounding).

Combined Strategies

Stratification combined with CV:

$$\hat{\mu}_t = \frac{1}{n_t}\sum_{i=1}^{n_t} X_{\mathrm{c,t,i}} = \frac{1}{n_t}\sum_{i=1}^{n_t}[X_{t,i} - \beta_t(A_{t,i} - a\, b_t)].$$

We minimize $\sigma_t^2 = Var[X_{\mathrm{c,t,i}}]$ by taking

$$\beta_t = \beta_t^* = \frac{Cov[A_{t,i},\ X_{t,i}]}{Var[A_{t,i}]} = \frac{Cov[A_i, X_i | B_i = b_t]}{(ab_t)}.$$

The addition of a CV alters the σ_t^2: the optimal allocation is no longer the same.

With $n_0 = 800$ pilot trials, we obtain $(\beta_1, \beta_2, \beta_3, \beta_4) = (1.020, 0.648, 0.224, -0.202)$ and $(n_1, n_2, n_3, n_4) = (131, 503, 247, 119)$ as estimates of the optimal values.

Combined Strategies

By repeating the experiment with $n = 100000$, we can find the following estimates for the variance ($\pm 1\%$):

$Var[X_n] = 21998$; $Var[X_{\mathrm{i,n}}] = 17996$; $Var[X_{\mathrm{c,n}}] = 3043$; $Var[X_{\mathrm{so,c,n}}] = 885$.

Numerical Results for $n = 1000$

Method	Estimator	Mean	$S_n^2(\pm 9\%)$	Ratio
Crude estimator	$\bar{X}_n$	1518.2	21615	1.000
Indirect	$\bar{X}_{\mathrm{i,n}}$	1502.5	18389	0.851
CV A_i, with pilot runs	$\bar{X}_{\mathrm{c,n}}$	1510.1	3305	0.153
CV A_i, no pilot runs	$\bar{X}_{\mathrm{ce,n}}$	1510.2	3310	0.153
Indirect + CV, no pilot runs	$\bar{X}_{\mathrm{i,c,n}}$	1510.1	3309	0.153
Stratification (proportional)	$\bar{X}_{\mathrm{sp,n}}$	1509.5	1778	0.082
Stratification (optimal)	$\bar{X}_{\mathrm{so,n}}$	1509.4	1568	0.073
Strat. (proportional) + CV	$\bar{X}_{\mathrm{sp,c,n}}$	1509.2	1140	0.053
Strat. (optimal) + CV	$\bar{X}_{\mathrm{so,c,n}}$	1508.3	900	0.042

Comparison of Two Similar Systems

Now suppose it is possible to slightly decrease the parameter γ of the distribution governing the service durations, by taking

$$\gamma = \gamma_2 = \gamma_1(1-\delta),$$

for $\delta > 0$ being very small.

If we use the same random numbers, this is equivalent to multiplying the service durations by $(1-\delta)$.

We want to estimate $\mu(\gamma_2) - \mu(\gamma_1) = \mathbb{E}_\gamma[X_2] - \mathbb{E}_\gamma[X_1]$, in order to measure the effect of slightly increasing the speed of the servers.

Comparison of Two Similar Systems

We simulate n days for each value of γ.

- $X_{1,i}$ = value of $G_i(s)$ with γ_1;
- $X_{2,i}$ = value of $G_i(s)$ with γ_2;
- $\Delta_i = X_{2,i} - X_{1,i}$,

$$\bar{\Delta}_n = \frac{1}{n} \sum_{i=1}^{n} \Delta_i$$

We can simulate $X_{1,i}$ and $X_{2,i}$

(i) with independent random variables (*IRN*),

(ii) with common random variables (*CRN*).

Comparison of Two Similar Systems

As

$$Var[\Delta_i] = Var[X_{1,i}] + Var[X_{2,i}] - 2Cov[X_{1,i}, X_{2,i}],$$

the goal of common random variables is to make the covariance positive.

How to implement CRNs?

Use different random sequences to generate:

1. The load factor B_i;
2. The inter-arrival times;
3. The call durations;
4. The patience durations.

Everything is generated by inversion: a uniform distribution is used for each random variable.

Synchronization

When service durations change, waiting times also change, and abandonment decisions may consequently change.

If we generate service durations only for clients who do not abandon, we may lose synchronization: we might have to generate fewer service durations in one system than in the other.

Service durations can be generated:

(a) for all calls, including those that abandon,

(b) only for served calls.

Synchronization

Similarly, the patience duration does not need to be generated for clients who do not wait. It can be generated:

(c) for all calls,

(d) only if necessary.

Synchronization: results (with $n = 10^4$)

Method	$\delta = 0.1$		$\delta = 0.01$		$\delta = 0.001$	
	$\bar{\Delta}_n$	$\widehat{Var}[\Delta_i]$	$\bar{\Delta}_n$	$\widehat{Var}[\Delta_i]$	$\bar{\Delta}_n$	$\widehat{Var}[\Delta_i]$
IRN (a + c)	55.2	56913	4.98	45164	0.66	44046
IRN (a + d)	52.2	54696	7.22	45192	-1.82	45022
IRN (b + c)	50.3	56919	9.98	44241	1.50	45383
IRN (b + d)	53.7	55222	5.82	44659	1.36	44493
CRN, no sync. (b + d)	56.0	3187	5.90	1204	0.19	726
CRN (a + c)	56.4	2154	6.29	37	0.62	1.8
CRN (a + d)	55.9	2161	6.08	158	0.74	53.8
CRN (b + c)	55.8	2333	6.25	104	0.63	7.9
CRN (b + d)	55.5	2323	6.44	143	0.59	35.3

Induction of Correlation

Sufficient conditions for $Cov[X, Y]$ to be positive or negative?

How to maximize or minimize correlation for given marginal distributions?

Theorem (Fréchet bounds)

Among pairs of random variables (X, Y) *with marginal distribution functions* F *and* G*, the pair* $(X, Y) = (F^{-1}(U), G^{-1}(U))$ *where* $U \sim U(0,1)$*, maximizes* $\rho[X, Y]$*, and the pair* $(X, Y) = (F^{-1}(U), G^{-1}(1-U))$ *minimizes* $\rho[X, Y]$*.*

Correlation Induction

For example, if the distribution function of a service duration is F in the first system and G in the second, and if we generate the service durations by $X = F^{-1}(U)$ and $Y = G^{-1}(U)$, then $Cov[X, Y] \geq 0$.

Theorem

Let $X = f(U)$ *where* $U \sim U(0,1)$ *and* $Y = g(V)$ *where* $V \sim U(0,1)$. *Then,*

- *if* f *and* g *are monotonic in the same direction and* $V = U$*, then* $Cov[X, Y] \geq 0$;
- *if* f *and* g *are monotonic in the same direction and* $V = 1 - U$*, then* $Cov[X, Y] \leq 0$.

Common Random Numbers (CRN)

We use $\Delta = X_2 - X_1$ to estimate $\mu_2 - \mu_1 = E[X_2] - E[X_1]$. Then

$$Var[\Delta] = Var[X_1] + Var[X_2] - 2\,Cov[X_1, X_2].$$

Goal: to induce a positive correlation between X_1 and X_2 without changing their individual distributions.

Technique: use the same random numbers to simulate both systems, trying to maintain *synchronization* as best as possible.

If $X_k = f_k(\boldsymbol{U}_k) = f_k(U_{k,1}, U_{k,2}, \ldots)$ for $k = 1, 2$, using RNGs everywhere means taking $\boldsymbol{U}_1 = \boldsymbol{U}_2$.

Common Random Numbers (CRN)

Theorem

If f_1 and f_2 are monotonic in the same direction with respect to all their arguments, then $Cov[X_1, X_2] \geq 0$.

To *maximize* the correlation, one should generate X_1 and X_2 directly through inversion!

Example: Lindley Process

$W_{i+1} = \max(0,\ W_i + S_i - A_i)$, where $S_i - A_i$ is independent of W_i.

If X_1 and X_2 are non-decreasing functions of the W_i for two Lindley processes simulated with common random numbers, then $Cov[X_1, X_2] \geq 0$.

Sometimes it is very difficult to check the conditions of the theorem.

For example, in a bank or a call center, monotonicity is hard to verify due to abandonment possibilities. Moreover, the conditions of the theorem are *sufficient*, but *not necessary*.

Common Random Numbers (CRN)

To *test* if it is effective in practice: conduct a *pilot experiment with common random numbers* and estimate $Cov(X_1, X_2)$.

It is not necessary to perform simulations without common random numbers: to estimate the variance one would have without common random numbers, it is sufficient to take the empirical version of $Var[X_1] + Var[X_2]$.